

ery good afternoon everyone. At your lunch
0:06 two answers I got you people are doin...

Revision Notes: Java Arrays & Scanner Class

1. Value vs Address Concept

- 1000 can be either a **value** or an **address** depending on the variable type
- If held by a normal variable → **value**
- If held by a reference variable → **address**

2. Java Naming Conventions

Convention	Description	Example	Usage
Camel Case	First word lowercase, subsequent words capitalized	makeSound, numberOfLegs	Variables, Methods
Pascal Case	Every word's first letter capitalized	StreetDog, DogClass	Class Names
Snake Case	Words separated by underscore	MAX_VALUE, my_variable	Constants
Kebab Case	Words separated by hyphen	my-variable	Not allowed in Java

3. Array Fundamentals

One-Dimensional (1D) Array

java

```
1 | int[] a = new int[5]; // Creates array with 5 elements
   | (indices 0-4)
```

Two-Dimensional (2D) Array

java

```
1 | int[][] a = new int[2][5]; // 2 classrooms, 5 students
   |     each
2 | // Row = classroom, Column = student
```

- **First bracket** → number of rows (classrooms)
- **Second bracket** → number of columns (students)

Three-Dimensional (3D) Array

java

```
1 | int[][][] a = new int[2][3][5]; // 2 schools, 3
   |     classrooms, 5 students
```

4. Array Indexing

- Index starts from 0 in Java
- Last index = `array.length - 1`
- Loop condition: `i < array.length` (NOT `<=`)

5. Scanner Class for User Input

Importing Scanner

java

```
1 | import java.util.Scanner;
```

Creating Scanner Object

java

```
1 | Scanner scan = new Scanner(System.in);
```

Input Methods

Method	Return Type	Purpose
nextInt()	int	Integer input
nextLong()	long	Long input
nextShort()	short	Short input
nextByte()	byte	Byte input
nextFloat()	float	Float input
nextDouble()	double	Double input
nextBoolean()	boolean	Boolean input
next()	String	Single word input
nextLine()	String	Full line input

6. Array Input/Output Using Loops

1D Array Example

java

```
1 // Input
2 for(int i = 0; i < n; i++) {
3     a[i] = scan.nextInt();
4 }
5
6 // Output
7 for(int i = 0; i < n; i++) {
8     System.out.print(a[i] + " ");
9 }
```

2D Array Example

java

```
1 // Input
2 for(int i = 0; i < classrooms; i++) {
3     for(int j = 0; j < students; j++) {
4         a[i][j] = scan.nextInt();
5     }
6 }
```

```
6     }
7
8     // Output
9     for(int i = 0; i < classrooms; i++) {
10         for(int j = 0; j < students; j++) {
11             System.out.print(a[i][j] + " ");
12         }
13         System.out.println(); // New line after each row
14     }
```

7. Regular vs Jagged Arrays

Type	Description	Example
Regular	Equal number of elements in each row	int[2][5] - both rows have 5 elements
Jagged	Different number of elements in each row	Row 1: 5, Row 2: 3

8. Key Interview Points

1. **Arrays are objects** in Java
 2. **Reference variable** stores address (hashcode)
 3. Default values: 0 for int, 0.0 for float/double, false for boolean, null for objects
 4. **Dimensionality** = number of pieces of information needed to access an element
 5. Always use loops with arrays
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9. Placement Tips (from TCS Interview)

- Focus on **projects** - most important
- Know your resume thoroughly
- Know at least **2 programming languages**
- Arrays & Strings are frequently asked topics
- AI-related questions are trending
- Practice aptitude from Preplnsta/TopAce